

EEE 304(January 2024)

Digital Electronics Laboratory

Final Project Report

Section: A2 Group: 02

Digital Logic Circuitry Based Smart Room Security and Automation system

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Academic Honesty Statement:

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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Table of Contents

1	Abstract.....	1
2	Introduction.....	1
3	Design	1
3.1	Problem Formulation (PO(b)).....	1
3.1.1	Identification of Scope	1
3.1.2	Literature Review.....	2
3.1.3	Formulation of Problem.....	2
3.1.4	Analysis	2
3.2	Design Method (PO(a)).....	3
4	Implementation	7
4.1	Description.....	9
5	Design Analysis and Evaluation	9
5.1	Novelty.....	9
5.2	Design Considerations (PO(c)).....	9
5.2.1	Considerations to public health and safety	9
5.2.2	Considerations to environment	9
5.2.3	Considerations to cultural and societal needs	10
5.3	Limitations of Tools (PO(e))	10
5.4	Ethical Issues (PO(h)).....	10
6	Reflection on Individual and Team work (PO(i)).....	11
6.1	Individual Contribution of Each Member.....	11
6.2	Mode of TeamWork.....	11
6.3	Diversity Statement of Team	11
6.4	Log Book of Project Implementation	12
7	Communication to External Stakeholders (PO(j))	12
7.1	Executive Summary.....	12
7.2	User Manual.....	13
7.3	YouTube Link.....	13
8	Project Management and Cost Analysis (PO(k)).....	13
9	Future Work (PO(l)).....	13
10.	References	14

1 Abstract

We are proposing a room security system based on digital logic circuits. The user will input a password to gain access which will be compared to the actual password stored in the system. If the entered password matches, access will be granted. But if wrong password is pressed, one will be given an opportunity of three times to enter password then if one fails, we will freeze the system for 15 seconds.

2 Introduction

To protect a highly secured area, traditional lock-and-key security is sometimes just not enough. Keys can get on the wrong hands. Those can be duplicated too. A good solution is to implement a password based lock. Only authorized personnel can unlock it. There is almost zero risk of incidents like key duplication and key-theft without the involvement of the authorized personnel themselves.

3 Design

3.1 Problem Formulation (PO(b))

Physical keys are most used technology to unlock a door. But it has some liabilities. Using physical keys has risk of safety as it can be stolen, doubled and provides poor safety. Password based smart door lock system can be a solution. It allows people have more security and lets a person unlock door without physical key.

3.1.1 Identification of Scope

- ✓ Utilizes fundamental principles of digital electronics, logic gates and integrated circuits to design a practical security and automation system.
- ✓ Demonstrates the ability to analyze security needs and design a system using digital logic circuits that ensures efficient room security and automation.
- ✓ Involves designing and implementing a complete system, showcasing the ability to create solutions based on logical components, fulfilling real-world requirements.
- ✓ Ensures the design meets safety standards by addressing security and automation concerns in a real-world setting, considering the well-being of the user.

3.1.2 Literature Review

- ✓ Previous studies highlight the use of basic logic gates in building automated systems for security and control applications.
- ✓ Research shows that digital logic-based systems are effective in enhancing security through automation, reducing human error and improving reliability.
- ✓ Several works demonstrate the role of integrated circuits in reducing system complexity and improving the compactness of security and automation systems.
- ✓ Literature discusses challenges such as power consumption, system integration and real-time performance in security and automation systems.

3.1.3 Formulation of Problem

The purpose of the project is to make a door-locking mechanism that will open the door when the correct password is entered. A smart lock with digital password can simply ensure the security of home, business, locker and other valuables. Only authorized people can access to the restricted area. So, we will make an indoor password setting block and outdoor locking system. Authorized people can set the password. Thus, we can solve our safety issues.

3.1.4 Analysis

The project requires analysis of the security needs, automation features and environmental factors to ensure the system meets user expectations. The design process involves the selection and configuration of basic logic gates to fulfill specific security functions like door locking, motion detection and alert systems.

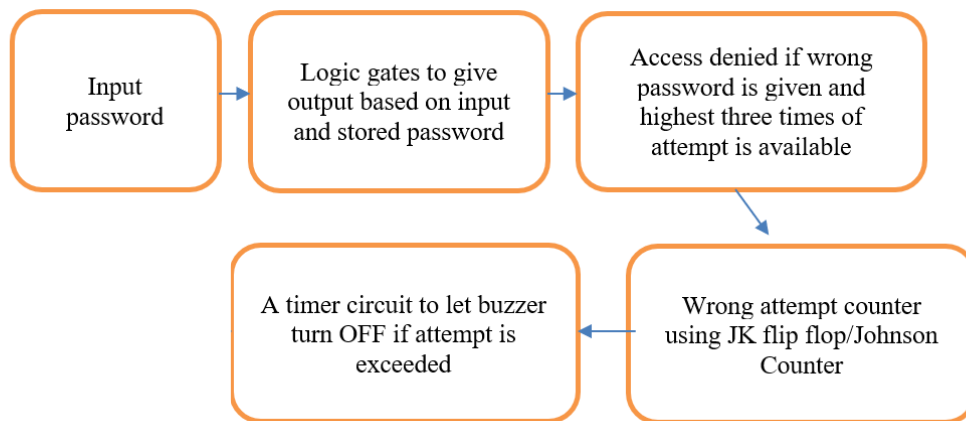
3.2 Design Method (PO(a))

We will use only digital logic ICs and gates to design our circuits. First of all, to input password we need mechanical pushbuttons for 0 to 9 numbers. We use encoder to convert into binary number. The binary numbers are passed through shift registers and input to the comparators. In the comparator block, the shift register output of both indoor and outdoor block are compared. If the password is same as preset password the door will unlock. Otherwise, consecutive several wrong attempts will freeze the system for a certain time interval. Then we can enter password again.

Component

1. 74 SERIES AND, OR, NOT, NAND, PRIORITY ENCODER
2. 4-BIT S/P SHIFT REGISTER
3. 4-BIT MAGNITUDE COMPARATOR
4. 40 SERIES BILATERAL SWITCH, JOHNSON COUNTER, BUFFER
5. RESISTOR, CAPACITOR
6. POTENTIOMETER, PUSH BUTTON
7. BREADBOARD
8. SG90 SERVO MOTOR
9. LED

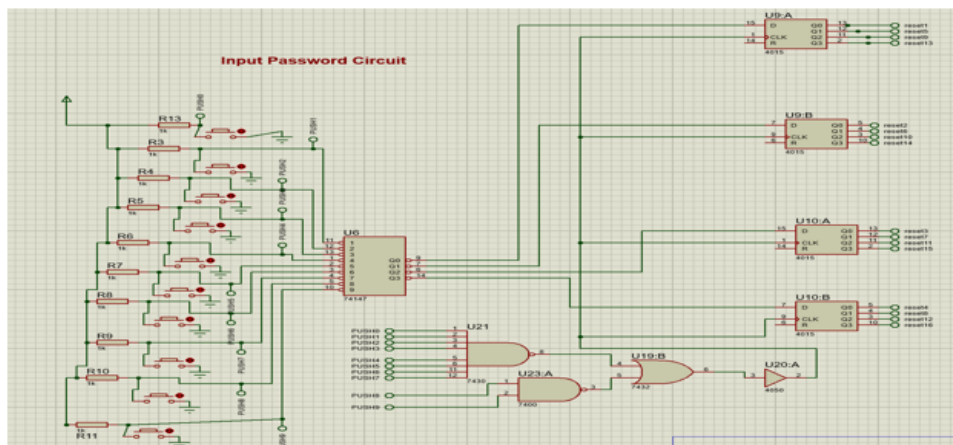
3.3 Circuit Diagram



3.4 Simulation Model

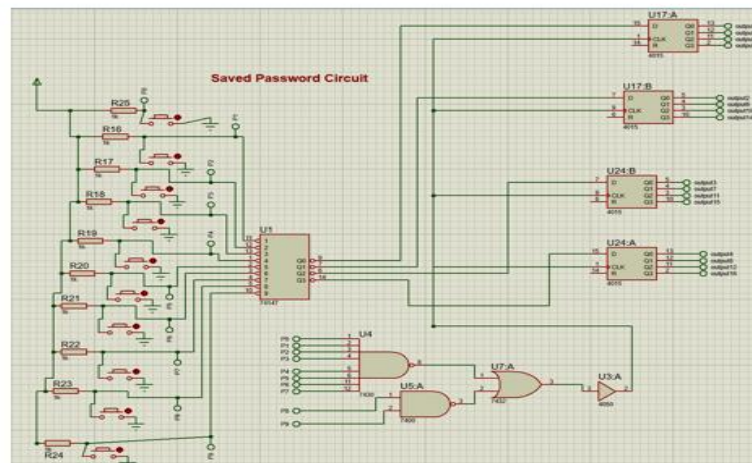
Input password region

Components: 74147 Encoder, 4015 Serial to Parallel Shift Register, Clock Circuit



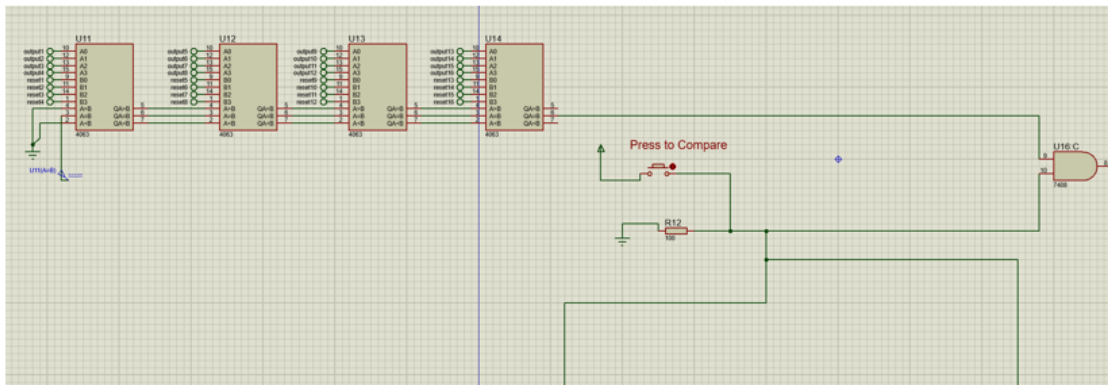
Saved password region

Components: 74147 Encoder, 4015 Serial to Parallel Shift Register, Clock Circuit



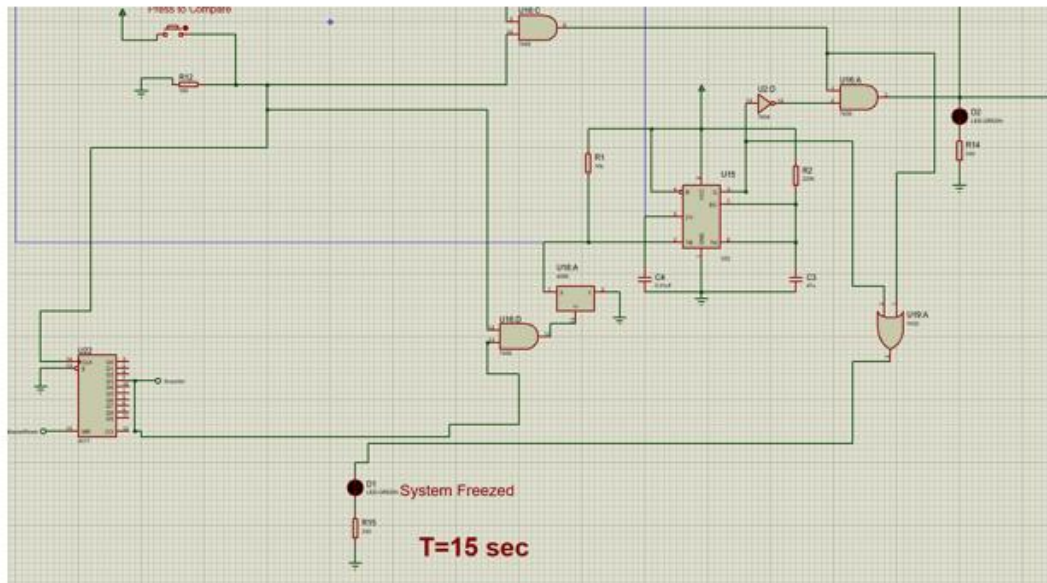
Comparator circuit

Components: 4063 comparator IC, Push button, 7408



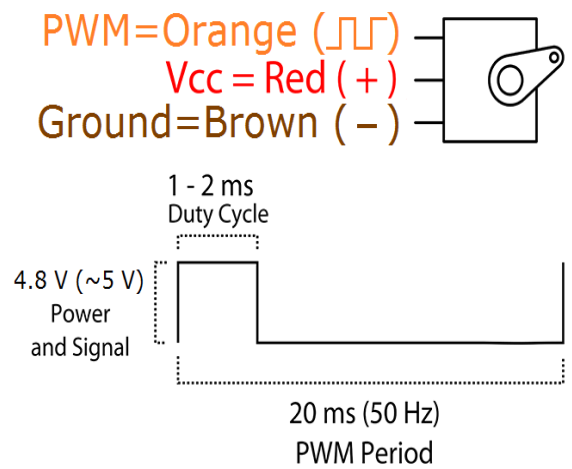
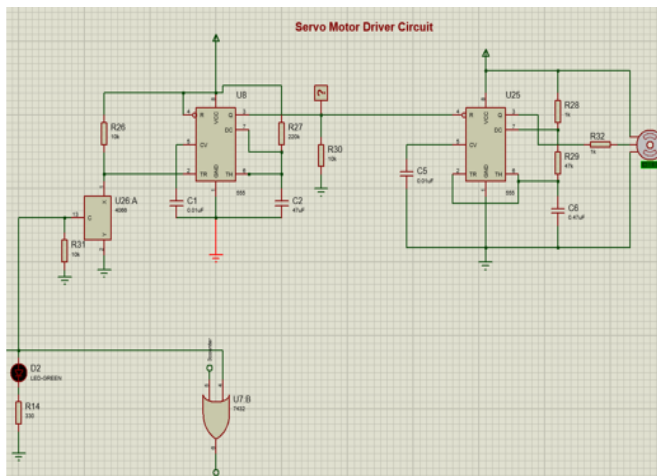
Counter and freezing Circuit:

Components: 4017 Johnson Counter, 555 Timer in Monostable Mode, 4066 Bilateral Switch

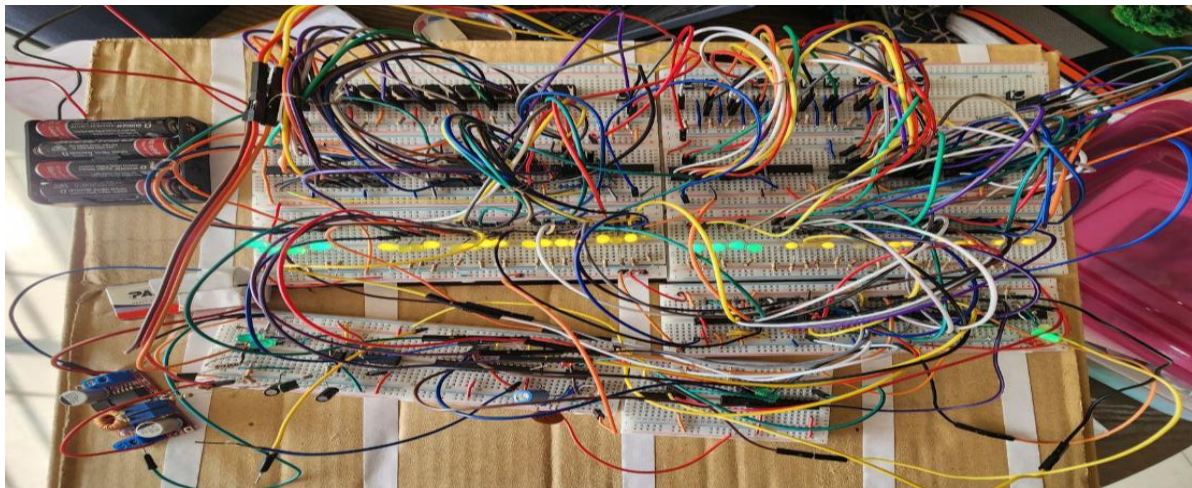


Servo Motor Driver Circuit :

Components: Two 555 timer in series (Monostable, Astable Mode)



3.5 Hardware Design

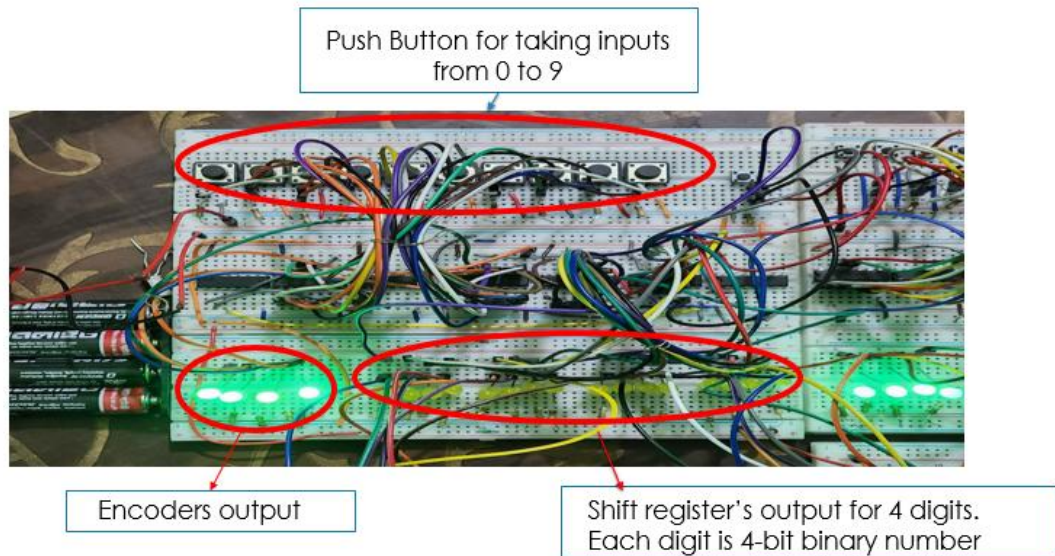


Servo motor associated with door lock to model a simple door lock mechanism

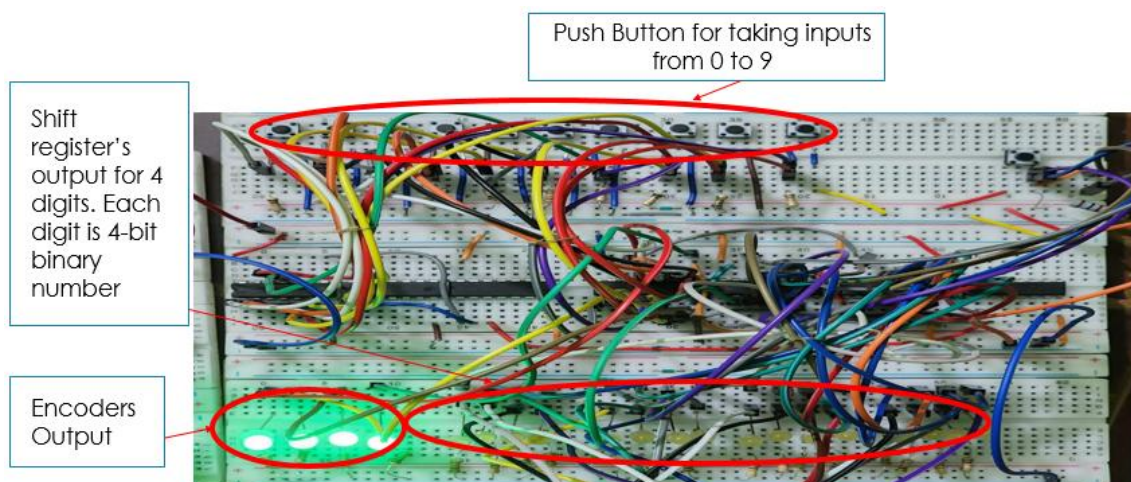


4 Implementation

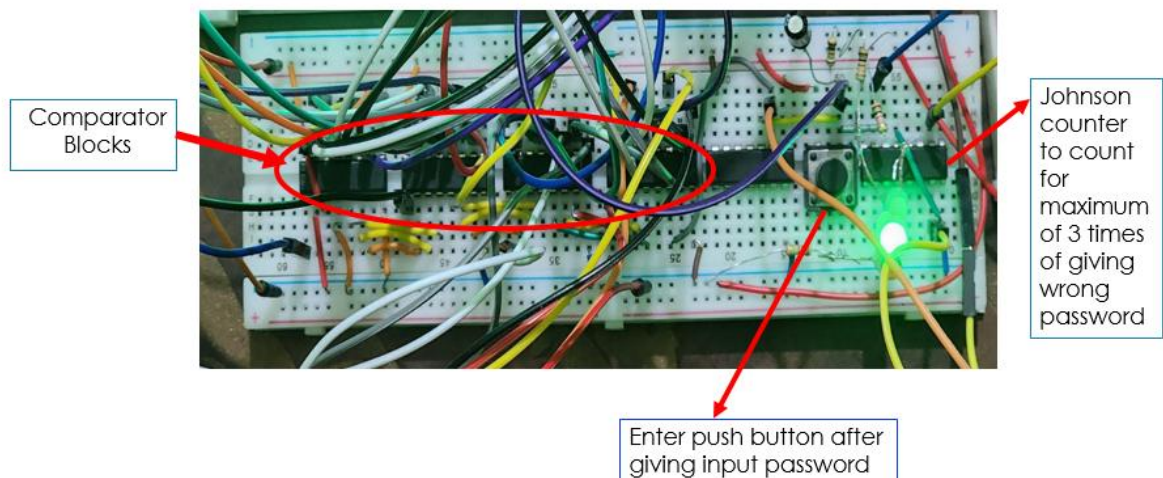
Input Password Region:



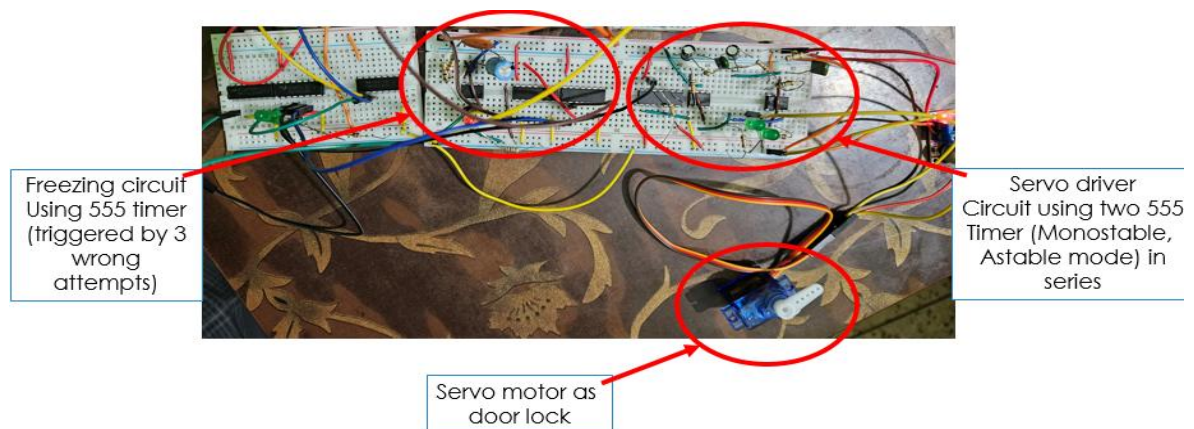
Saved password Region:



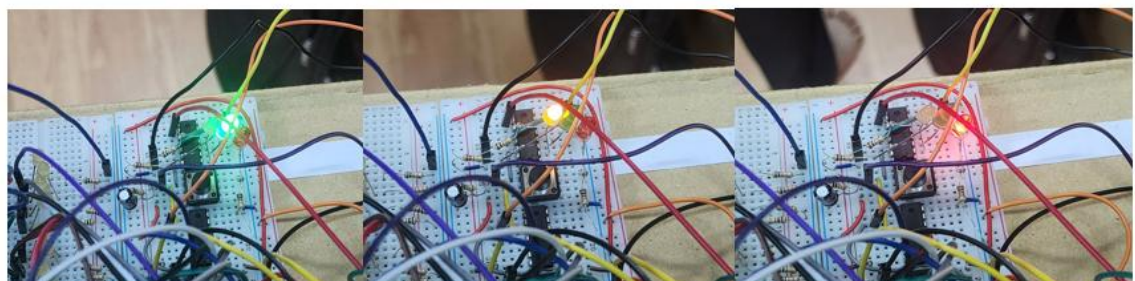
Comparator blocks and Johnson Counter:



Servo motor driver and freezing circuit



Wrong attempt counter:



Wrong Attempt Counter
Green – 1st try
Yellow – 2nd try
Red-3rd (last try)

4.1 Description

We have indoor and outdoor block. Firstly, in the indoor block, we set a 4-bit password. Mechanical switch may have some issues that sometimes if we push once, push button mechanical things cannot work properly and our one-bit input can be 2bit same binary number. If this problem happens, we can observe LED outputs to see the entered number. After setting the password, we want demonstrate it. When we entered the same 4-bit input as password, the output of 4 shift registers of each block is compared using comparator. If comparator output is equal, then the door lock driver works and unlock the door. If multiple wrong attempts have entered, using Johnson Decade counter, the system counts and system freezes. After some time, we can enter password again.

5 Design Analysis and Evaluation

5.1 Novelty

- ✓ Our project introduces a novel approach to password-based door locking systems by leveraging digital logic circuits for enhanced reliability, speed, and cost-efficiency.
- ✓ Unlike traditional microcontroller-based designs, our system uses combinational and sequential logic to process password inputs from a pushbutton, ensuring minimal power consumption, faster response times, and increased security against software vulnerabilities.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

- ✓ Traditional Lock & Key is an integral part of our day-to-day security and safety. We have tried to take it to the next level by implementing a Digital Electronics based solution.
- ✓ Our solution is free from the risk of key-theft, key-copying, poor security, and brute force password cracking which add extra layers of security.

5.2.2 Considerations to environment

- ✓ We did not use any components in our project that is directly harmful to the environment.
- ✓ Each component used in our project is widely available, so mass producing our

system will not cause any shortage of critical and rarely available materials.

- ✓ Our system does not cause any emission of harmful substances.

5.2.3 Considerations to cultural and societal needs

- ✓ With the ongoing digitalization of our society, compatibility issues of the very basic day-to-day tools will arise.
- ✓ If we want to setup a smart home, the door locks need to be controlled from digital signal. With traditional door locks, it is not possible. But our system can make it happen.

5.3 Limitations of Tools (PO(e))

Shift register: Shift register was not working properly when we used two shift registers for 4 encoder outputs. So, we used 1 shift register for each encoder output and the problem was solved using more ICs.

Mechanical Push Button: We used mechanical pushbuttons as keypad for entering password. Sometimes, for one push we get 2 consecutive same binary input.

Delay: The switches and components are not ideal hence they have delay that sometimes cause problem in the output when opening the lock.

5.4 Ethical Issues (PO(h))

The project incorporates ethical principles by prioritizing user privacy, ensuring secure access control, and designing the system to prevent misuse or unauthorized tampering. During development, one ethical challenge was balancing cost-effectiveness with reliability to ensure the system remained accessible to diverse socioeconomic groups without compromising quality. This was mitigated by carefully selecting components that met safety and performance standards while maintaining affordability. Additionally, the design process included thorough testing to ensure the system adhered to safety regulations, avoiding potential harm to users or the environment. These measures reflect a commitment to fairness, transparency, and responsibility in addressing ethical concerns.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

- Resource collection: 2006048
- Simulation: 2006048,2006065
- Input and set password circuit: 2006036,2006060
- Shift register & Clock circuit: 2006048,2006060
- Comparator circuit: 2006036,2006060
- Freezing circuit: 2006036,2006065
- Servo driver circuit: 2006060,2006065

6.2 Mode of Teamwork

Our combined effort consisted of planning, setting up goals, working towards goals, and adjusting goals based on progress. Although we were connected through online platforms all the time, we sat together in person on weekends to brainstorm ideas, to discuss next steps, and to make circuit connections.

6.3 Diversity Statement of Team

The four teammates are divided into two subgroups and each subgroup specializes in different areas than the other. The group of Emam Al Mahdee and S.M. Mahfuz mainly focuses on coming up with a concept, doing research on it, creating initial prototype, and simulating it and buying necessary components from the market. The group of Md. Sajjad Hossain Mridul and Md. Shakib Hossain focuses on final implementation of the design, which includes design verification via simulation, making physical circuit connections and debugging the system.

6.4 Log Book of Project Implementation

Date	Milestone achieved	Individual Role	Team Role	Comments
20-09-2024	Resource collection	2006048		
22-09-2024	Project Proposal	All		Project Accepted
26-09-2024	Buying component	All		
02-10-2024	Simulation	2006048,2006065		Done successfully
03-11-2024	Project Update	All		
02-12-2024	Hardware(Input and preset password	2006036,2006060		
04-12-2024	Hardware(Shift register and clock circuit)	2006048,2006060		At first shift register was not working but after adding buffer it was performing
05-12-2024	Hardware(comparator Circuit)	2006036,2006060		
13-12-2024	Hardware(freezing circuit)	2006036,2006065		
14-12-2024	Servo driver circuit	2006060,2006065		
15-12-2024	Project Demonstration	All		Executed successfully

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

The project is a password-based door lock system that aims to replace the traditional lock and key system. The system has 2 sets of switches from 0 to 9. One set of switches is placed inside the house or secured place (Indoor) and the other side of switch is placed in front of the door (Outdoor). The password, a 4-digit number is set in the Indoor side and then the intended user has to input the password in the Outdoor switch. If the two passwords match, then the door will open, If the passwords do not match for 3 times, then the lock will be frozen that is the door won't open despite correct password and after a certain time the door lock can be opened again.

7.2 User Manual

The using manual for the project is quite simple and intuitive.

1. Set the 4-digit password of the Indoor block
2. Enter the password in the Outdoor block.

If the two passwords are the same then the lock will open. If the password is incorrect more than 3 times the lock will be frozen for 15 seconds, during that time even if the correct password is entered the lock will not open.

7.3 YouTube Link

Here is our project's YouTube link:

<https://youtu.be/eb5XoA1jTqU?si=Lr-8mZzssOTJyLNs>

8 Project Management and Cost Analysis (PO(k))

Name of Component	Cost(Taka)
74147(9 to 4 priority encoder)	250
7400(2 input NAND gate)	250
7430(8 Input NAND gate)	250
4017(Johnson Counter)	250
4015 shift register	250
PVC board +accessories	300
7408(AND gate)	250
7404(NOT gate)	250
7432(OR gate)	250
4066(Bilateral switch)	250
4050(Buffer)	250
SG90 servo motor	120
4063 (4 bit magnitude comparator)	100
Bread Board	1600
Jumper Cable	500
LED	600
Flat wire	300
Total	6020

9 Future Work (PO(l))

We can set up the door lock system in any smart home system. That'll be totally authorized. The mechanical switches can be debounced that would solve the problem of switching. The memories can be changed to something more permeant such as DRAM or ROM etc. The power source can be made more robust.

10. References

1. <https://www.instructables.com/Digital-Door-lock-System-With-Keypad/>
2. https://www.researchgate.net/publication/355348268_Hex-Password_Based_Lock_Security_System
3. <https://www.alldatasheet.com/>